

TEST CASE EXECUTION TEMPLATE

Voice Interaction (STT/TTS) - Test Case 1 of 4

TEST CASE TITLE	TC-VOICE-01: Voice command correctly recognised and answered (positive)	TEST DATE	2026-06-27
TEST SCENARIO	Speech-to-text recognition of a spoken command, with a spoken (TTS) response.	TEST DESIGNED BY	SafeNav team
TEST DESCRIPTION	The user taps to talk and speaks an assistant command ("list places"). The system recognises the intent and gives the correct spoken response. An assistant-only command is used so this case verifies the voice subsystem itself without overlapping the navigation or obstacle-detection test suites.		
PRE-CONDITIONS	- App opened on the user (voice) screen; state is idle. - Microphone permission granted; environment quiet; media volume audible. - Internet available and Azure Speech reachable.		

STEP ID	STEP DESCRIPTION	EXPECTED RESULTS	ACTUAL RESULTS	PASS / FAIL	ADDITIONAL NOTES
1	Tap anywhere on the screen and, after the cue, say "list places".	An activation cue plays and the app enters the LISTENING state; the full phrase is captured.	Works as intended (matches the expected result).	Pass	Tap-to-talk (toggle); audio is captured from the moment of the tap.
2	Finish speaking and wait briefly.	The recognised transcript is shown as a "You said" caption and the app moves to processing.	Works as intended (matches the expected result).	Pass	Recognition tolerates minor mispronunciation via fuzzy matching (e.g. "list place" still maps to the command).
3	Listen to the spoken response.	The assistant speaks the available locations (the faculties, the library, and any cafeterias), shows it as an "Assistant" caption, then returns to idle.	Works as intended (matches the expected result).	Pass	

TEST CASE EXECUTION TEMPLATE

Voice Interaction (STT/TTS) - Test Case 2 of 4

TEST CASE TITLE	TC-VOICE-02: Unclear or unsupported speech handled safely (negative)	TEST DATE	2026-06-27
TEST SCENARIO	The system must not execute a wrong command when the speech is unsupported or ambiguous.	TEST DESIGNED BY	SafeNav team
TEST DESCRIPTION	For unrelated or out-of-vocabulary speech the assistant states it did not understand and performs no action. For input that is close to two different commands (genuinely ambiguous), the system does not force a guess. No incorrect command is executed.		
PRE-CONDITIONS	- App opened on the user (voice) screen; state is idle. - Microphone permission granted; environment quiet; media volume audible. - Internet available and Azure Speech reachable.		

STEP ID	STEP DESCRIPTION	EXPECTED RESULTS	ACTUAL RESULTS	PASS / FAIL	ADDITIONAL NOTES
1	Tap and say an unrelated phrase (e.g. "what the hell").	The recognised text may appear, but NO command is triggered.	Works as intended (matches the expected result).	Pass	False-positive guard: casual/unrelated speech does not launch a command.
2	Listen to the response.	The assistant says "Sorry, I didn't understand that" (prefixed with the user's name if set) and stays idle; nothing is started or changed.	Works as intended (matches the expected result).	Pass	
3	Tap and say an ambiguous near-command (e.g. "start addition").	The system does not force-match it to a command; it is treated as not understood and no action is taken.	Works as intended (matches the expected result).	Pass	Ambiguous input that is roughly equidistant between commands is intentionally not guessed - important for a safety app.

TEST CASE EXECUTION TEMPLATE

Voice Interaction (STT/TTS) - Test Case 3 of 4

TEST CASE TITLE	TC-VOICE-03: Spoken-feedback management - no spam and repeat (positive)	TEST DATE	2026-06-27
TEST SCENARIO	The shared speech output suppresses repeated guidance and can replay the last message.	TEST DESIGNED BY	SafeNav team
TEST DESCRIPTION	While guidance is being produced, the same warning is announced once and then suppressed for a cooldown period - even as it persists or its distance changes - so it does not spam the user. Saying "repeat" replays the last thing the user heard.		
PRE-CONDITIONS	- Obstacle detection is active (or guidance is otherwise being produced). - Media volume audible; environment quiet. - Microphone permission granted.		

STEP ID	STEP DESCRIPTION	EXPECTED RESULTS	ACTUAL RESULTS	PASS / FAIL	ADDITIONAL NOTES
1	Keep the same obstacle in view (e.g. a chair ahead) for about 10-15 seconds.	The warning (e.g. "chair ... ahead") is spoken once, then NOT repeated every frame; it may repeat only after the cooldown elapses.	Works as intended (matches the expected result).	Pass	Cooldown is keyed on obstacle type + region, independent of distance.
2	Let only the distance to that obstacle change (e.g. step closer).	A distance-only change does NOT re-announce the warning (treated as the same warning).	Works as intended (matches the expected result).	Pass	A different obstacle, or the same obstacle in a different region, IS announced immediately.
3	Tap and say "repeat".	The assistant replays the last message the user heard (the last guidance line or reply), shown as a caption.	Works as intended (matches the expected result).	Pass	"Repeat" is global: it covers navigation, obstacle, and assistant messages - not only the last assistant reply.

TEST CASE EXECUTION TEMPLATE

Voice Interaction (STT/TTS) - Test Case 4 of 4

TEST CASE TITLE	TC-VOICE-04: Tap-to-talk toggle and barge-in (positive)	TEST DATE	2026-06-27
TEST SCENARIO	One consistent tap rule controls listening and can interrupt the assistant's speech.	TEST DESIGNED BY	SafeNav team
TEST DESCRIPTION	A tap starts listening (with a cue). Tapping while the assistant is speaking interrupts it and starts listening. Tapping again while listening cancels back to idle without running any command. This validates the single, consistent tap-to-talk (toggle) interaction.		
PRE-CONDITIONS	- App opened on the user (voice) screen. - Microphone permission granted; media volume audible; environment quiet.		

STEP ID	STEP DESCRIPTION	EXPECTED RESULTS	ACTUAL RESULTS	PASS / FAIL	ADDITIONAL NOTES
1	From idle, tap anywhere on the screen.	An activation cue plays and the app enters the LISTENING state (microphone open).	Works as intended (matches the expected result).	Pass	The whole screen is the tap target.
2	Make the assistant speak (e.g. say "more info"); while it is speaking, tap the screen.	The speech stops immediately and the app enters LISTENING (it does not wait for the message to finish).	Works as intended (matches the expected result).	Pass	Barge-in works the same whether interrupting an assistant reply or guidance.
3	While listening, tap again without speaking.	A short stop/"off" cue plays and the app returns to idle; no command is executed.	Works as intended (matches the expected result).	Pass	Tap-to-talk is a toggle (tap to start / interrupt; tap again to cancel) - not press-and-hold.